The notion of a "gauge" is introduced in Section 5.5 and is used to provide an alternative method of proving the fundamental properties of continuous functions. The main significance of this concept, however, is in the area of integration theory where gauges are essential in defining the generalized Riemann integral. This will be discussed in Chapter 10.

Monotone functions are an important class of functions with strong continuity properties and they are discussed in Section 5.6.

Section 5.1 Continuous Functions

In this section, which is very similar to Section 4.1, we will define what it means to say that a function is continuous at a point, or on a set. This notion of continuity is one of the central concepts of mathematical analysis, and it will be used in almost all of the following material in this book. Consequently, it is essential that the reader master it.

5.1.1 Definition Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is **continuous at** c if, given any number $\varepsilon > 0$, there exists $\delta > 0$ such that if x is any point of A satisfying $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

If f fails to be continuous at c, then we say that f is **discontinuous at** c.

As with the definition of limit, the definition of continuity at a point can be formulated very nicely in terms of neighborhoods. This is done in the next result. We leave the verification as an important exercise for the reader. See Figure 5.1.1.

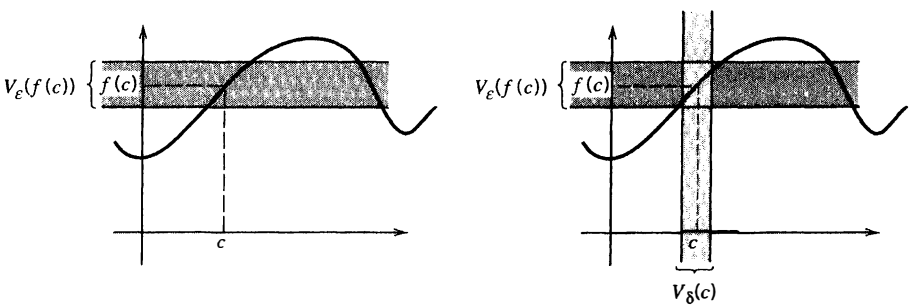


Figure 5.1.1 Given $V_\varepsilon(f(c))$, a neighborhood $V_\delta(c)$ is to be determined

5.1.2 Theorem *A function $f : A \to \mathbb{R}$ is continuous at a point $c \in A$ if and only if given any ε-neighborhood $V_\varepsilon(f(c))$ of $f(c)$ there exists a δ-neighborhood $V_\delta(c)$ of c such that if x is any point of $A \cap V_\delta(c)$, then $f(x)$ belongs to $V_\varepsilon(f(c))$, that is,*

$$f(A \cap V_\delta(c)) \subseteq V_\varepsilon(f(c)).$$

Remarks (1) If $c \in A$ is a cluster point of A, then a comparison of Definitions 4.1.4 and 5.1.1 show that f is continuous at c if and only if

$$f(c) = \lim_{x \to c} f(x). \tag{1}$$

Thus, if c is a cluster point of A, then three conditions must hold for f to be continuous at c:

(i) f must be defined at c (so that $f(c)$ makes sense),

(ii) the limit of f at c must exist in $\mathbb{R}$ (so that $\lim_{x\to c} f(x)$ makes sense), and

(iii) these two values must be equal.

(2) If $c \in A$ is not a cluster point of A, then there exists a neighborhood $V_\delta(c)$ of c such that $A \cap V_\delta(c) = \{c\}$. Thus we conclude that a function f is automatically continuous at a point $c \in A$ that is not a cluster point of A. Such points are often called "isolated points" of A. They are of little practical interest to us, since they have no relation to a limiting process. Since continuity is automatic for such points, we generally test for continuity only at cluster points. Thus we regard condition (1) as being characteristic for continuity at c.

A slight modification of the proof of Theorem 4.1.8 for limits yields the following sequential version of continuity at a point.

5.1.3 Sequential Criterion for Continuity *A function $f : A \to \mathbb{R}$ is continuous at the point $c \in A$ if and only if for every sequence (x_n) in A that converges to c, the sequence $(f(x_n))$ converges to $f(c)$.*

The following Discontinuity Criterion is a consequence of the last theorem. It should be compared with the Divergence Criterion 4.1.9(a) with $L = f(c)$. Its proof should be written out in detail by the reader.

5.1.4 Discontinuity Criterion *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. Then f is discontinuous at c if and only if there exists a sequence (x_n) in A such that (x_n) converges to c, but the sequence $(f(x_n))$ does not converge to $f(c)$.*

So far we have discussed continuity at a *point*. To talk about the continuity of a function on a *set*, we will simply require that the function be continuous at each point of the set. We state this formally in the next definition.

5.1.5 Definition Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If B is a subset of A, we say that f is **continuous on the set** B if f is continuous at every point of B.

5.1.6 Examples **(a)** The constant function $f(x) := b$ is continuous on $\mathbb{R}$.

It was seen in Example 4.1.7(a) that if $c \in \mathbb{R}$, then $\lim_{x\to c} f(x) = b$. Since $f(c) = b$, we have $\lim_{x\to c} f(x) = f(c)$, and thus f is continuous at every point $c \in \mathbb{R}$. Therefore f is continuous on $\mathbb{R}$.

(b) $g(x) := x$ is continuous on $\mathbb{R}$.

It was seen in Example 4.1.7(b) that if $c \in \mathbb{R}$, then we have $\lim_{x\to c} g = c$. Since $g(c) = c$, then g is continuous at every point $c \in \mathbb{R}$. Thus g is continuous on $\mathbb{R}$.

(c) $h(x) := x^2$ is continuous on $\mathbb{R}$.

It was seen in Example 4.1.7(c) that if $c \in \mathbb{R}$, then we have $\lim_{x\to c} h = c^2$. Since $h(c) = c^2$, then h is continuous at every point $c \in \mathbb{R}$. Thus h is continuous on $\mathbb{R}$.

(d) $\varphi(x) := 1/x$ is continuous on $A := \{x \in \mathbb{R} : x > 0\}$.

It was seen in Example 4.1.7(d) that if $c \in A$, then we have $\lim_{x\to c} \varphi = 1/c$. Since $\varphi(c) = 1/c$, this shows that φ is continuous at every point $c \in A$. Thus φ is continuous on A.

(e) $\varphi(x) := 1/x$ is not continuous at $x = 0$.

Indeed, if $\varphi(x) = 1/x$ for $x > 0$, then φ is not defined for $x = 0$, so it cannot be continuous there. Alternatively, it was seen in Example 4.1.10(a) that $\lim_{x\to 0} \varphi$ does not exist in $\mathbb{R}$, so φ cannot be continuous at $x = 0$.

(f) The signum function sgn is not continuous at 0.

The signum function was defined in Example 4.1.10(b), where it was also shown that $\lim_{x\to 0} \operatorname{sgn}(x)$ does not exist in $\mathbb{R}$. Therefore sgn is not continuous at $x = 0$ (even though sgn 0 is defined). It is an exercise to show that sgn is continuous at every point $c \neq 0$.

Note In the next two examples, we introduce functions that played a significant role in the development of real analysis. Discontinuities are emphasized and it is not possible to graph either of them satisfactorily. The intuitive idea of drawing a curve in the plane to represent a function simply does not apply, and plotting a handful of points gives only a hint of their character. In the nineteenth century, these functions clearly demonstrated the need for a precise and rigorous treatment of the basic concepts of analysis. They will reappear in later sections.

(g) Let $A := \mathbb{R}$ and let f be Dirichlet's "discontinuous function" defined by

$$f(x) := \begin{cases} 1 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

We claim that f is *not continuous at any point of* $\mathbb{R}$. (This function was introduced in 1829 by P. G. L. Dirichlet.)

Indeed, if c is a rational number, let (x_n) be a sequence of irrational numbers that converges to c. (Corollary 2.4.9 to the Density Theorem 2.4.8 assures us that such a sequence does exist.) Since $f(x_n) = 0$ for all $n \in \mathbb{N}$, we have $\lim(f(x_n)) = 0$, while $f(c) = 1$. Therefore f is not continuous at the rational number c.

On the other hand, if b is an irrational number, let (y_n) be a sequence of rational numbers that converge to b. (The Density Theorem 2.4.8 assures us that such a sequence does exist.) Since $f(y_n) = 1$ for all $n \in \mathbb{N}$, we have $\lim(f(y_n)) = 1$, while $f(b) = 0$. Therefore f is not continuous at the irrational number b.

Since every real number is either rational or irrational, we deduce that f is not continuous at any point in $\mathbb{R}$.

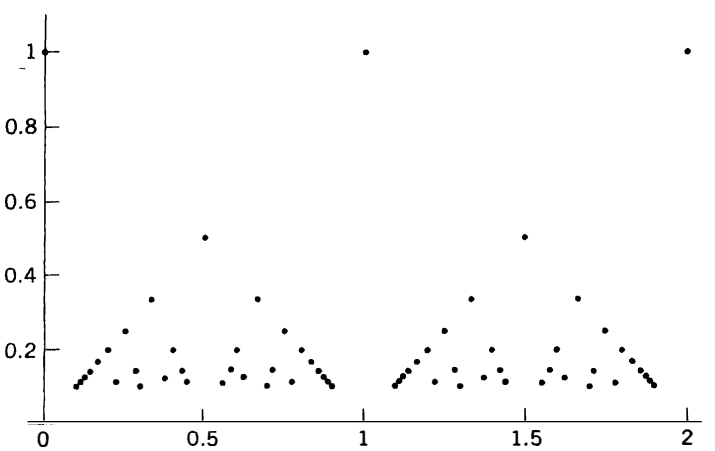


Figure 5.1.2 Thomae's function

(h) Let $A := \{x \in \mathbb{R}: x > 0\}$. For any irrational number $x > 0$ we define $h(x) := 0$. For a rational number in A of the form m/n, with natural numbers m, n having no common factors except 1, we define $h(m/n) := 1/n$. (We also define $h(0) := 1$.)

We claim that *h is continuous at every irrational number in A, and is discontinuous at every rational number in A.* (This function was introduced in 1875 by K. J. Thomae.)

Indeed, if $a > 0$ is rational, let (x_n) be a sequence of irrational numbers in A that converges to a. Then $\lim(h(x_n)) = 0$, while $h(a) > 0$. Hence h is discontinuous at a.

On the other hand, if b is an irrational number and $\varepsilon > 0$, then (by the Archimedean Property) there is a natural number n_0 such that $1/n_0 < \varepsilon$. There are only a finite number of rationals with denominator less than n_0 in the interval $(b-1, b+1)$. (Why?) Hence $\delta > 0$ can be chosen so small that the neighborhood $(b-\delta, b+\delta)$ contains no rational numbers with denominator less than n_0. It then follows that for $|x-b| < \delta, x \in A$, we have $|h(x) - h(b)| = |h(x)| \leq 1/n_0 < \varepsilon$. Thus h is continuous at the irrational number b.

Consequently, we deduce that Thomae's function h is continuous precisely at the irrational points in A. (See Figure 5.1.2.) □

5.1.7 Remarks **(a)** Sometimes a function $f: A \to \mathbb{R}$ is not continuous at a point c because it is not defined at this point. However, if the function f has a limit L at the point c and if we define F on $A \cup \{c\} \to \mathbb{R}$ by

$$F(x) := \begin{cases} L & \text{for} \quad x = c, \\ f(x) & \text{for} \quad x \in A, \end{cases}$$

then F is continuous at c. To see this, one needs to check that $\lim_{x\to c} F = L$, but this follows (why?), since $\lim_{x\to c} f = L$.

(b) If a function $g: A \to \mathbb{R}$ does not have a limit at c, then there is no way that we can obtain a function $G: A \cup \{c\} \to \mathbb{R}$ that is continuous at c by defining

$$G(x) := \begin{cases} C & \text{for} \quad x = c, \\ g(x) & \text{for} \quad x \in A. \end{cases}$$

To see this, observe that if $\lim_{x\to c} G$ exists and equals C, then $\lim_{x\to c} g$ must also exist and equal C.

5.1.8 Examples **(a)** The function $g(x) := \sin(1/x)$ for $x \neq 0$ (see Figure 4.1.3) does not have a limit at $x = 0$ (see Example 4.1.10(c)). Thus there is no value that we can assign at $x = 0$ to obtain a continuous extension of g at $x = 0$.

(b) Let $f(x) := x\sin(1/x)$ for $x \neq 0$. (See Figure 5.1.3.) It was seen in Example 4.2.8(f) that $\lim_{x\to 0}(x\sin(1/x)) = 0$. Therefore it follows from Remark 5.1.7(a) that if we define $F: \mathbb{R} \to \mathbb{R}$ by

$$F(x) := \begin{cases} 0 & \text{for} \quad x = 0, \\ x\sin(1/x) & \text{for} \quad x \neq 0, \end{cases}$$

then F is continuous at $x = 0$.

□

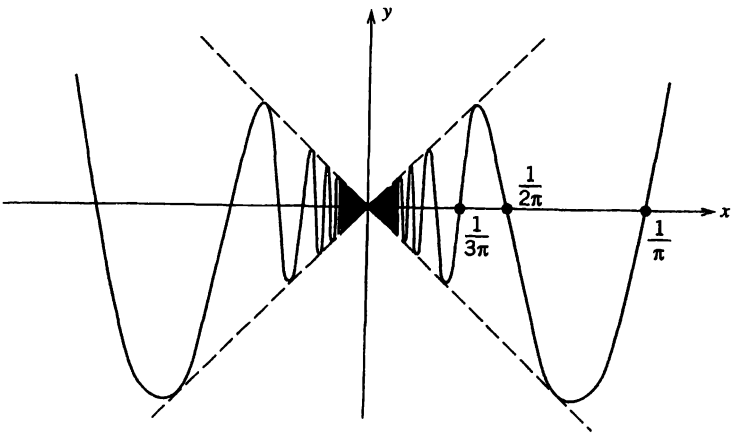


Figure 5.1.3 Graph of $f(x) = x \sin(1/x) \quad (x \neq 0)$

Exercises for Section 5.1

1. Prove the Sequential Criterion 5.1.3.
2. Establish the Discontinuity Criterion 5.1.4.
3. Let $a < b < c$. Suppose that f is continuous on $[a, b]$, that g is continuous on $[b, c]$, and that $f(b) = g(b)$. Define h on $[a, c]$ by $h(x) := f(x)$ for $x \in [a, b]$ and $h(x) := g(x)$ for $x \in [b, c]$. Prove that h is continuous on $[a, c]$.
4. If $x \in \mathbb{R}$, we define $[\![x]\!]$ to be the greatest integer $n \in \mathbb{Z}$ such that $n \leq x$. (Thus, for example, $[\![8.3]\!] = 8, [\![\pi]\!] = 3, [\![-\pi]\!] = -4$.) The function $x \mapsto [\![x]\!]$ is called the **greatest integer function.** Determine the points of continuity of the following functions:
   (a) $f(x) := [\![x]\!]$, (b) $g(x) := x[\![x]\!]$,
   (c) $h(x) := [\![\sin x]\!]$, (d) $k(x) := [\![1/x]\!] \quad (x \neq 0)$.
5. Let f be defined for all $x \in \mathbb{R}, x \neq 2$, by $f(x) = (x^2 + x - 6)/(x - 2)$. Can f be defined at $x = 2$ in such a way that f is continuous at this point?
6. Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$ be continuous at a point $c \in A$. Show that for any $\varepsilon > 0$, there exists a neighborhood $V_\delta(c)$ of c such that if $x, y \in A \cap V_\delta(c)$, then $|f(x) - f(y)| < \varepsilon$.
7. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous at c and let $f(c) > 0$. Show that there exists a neighborhood $V_\delta(c)$ of c such that if $x \in V_\delta(c)$, then $f(x) > 0$.
8. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous on $\mathbb{R}$ and let $S := \{x \in \mathbb{R} : f(x) = 0\}$ be the "zero set" of f. If (x_n) is in S and $x = \lim(x_n)$, show that $x \in S$.
9. Let $A \subseteq B \subseteq \mathbb{R}$, let $f : B \to \mathbb{R}$ and let g be the restriction of f to A (that is, $g(x) = f(x)$ for $x \in A$).
   (a) If f is continuous at $c \in A$, show that g is continuous at c.
   (b) Show by example that if g is continuous at c, it need not follow that f is continuous at c.
10. Show that the absolute value function $f(x) := |x|$ is continuous at every point $c \in \mathbb{R}$.
11. Let $K > 0$ and let $f : \mathbb{R} \to \mathbb{R}$ satisfy the condition $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in \mathbb{R}$. Show that f is continuous at every point $c \in \mathbb{R}$.
12. Suppose that $f : \mathbb{R} \to \mathbb{R}$ is continuous on $\mathbb{R}$ and that $f(r) = 0$ for every rational number r. Prove that $f(x) = 0$ for all $x \in \mathbb{R}$.
13. Define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) := 2x$ for x rational, and $g(x) := x + 3$ for x irrational. Find all points at which g is continuous.